ExirASO

Chemical Industries

A Responsible Business



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ExirASO
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We Act Responsibly

LABSA, SLES, OleoChemicals And Other Surfactants Raw Material Producer



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About us

Who Are We?

Exir **ASO** is a company located in Soltanieh Industrial Zone 45 km away from Zanjan city, Iran. It produces a variety of surfactants and oleochemicals.

The company's slogan is **"Innovation Is Our Business Plan"** and it always seeks to implement the latest standards and technologies. The organization is composed of a highly motivated and professional team that has made the culture of responsiveness, innovation, sustainability and customer orientation - our fundamental principles. We believe that the winning businesses of tomorrow will be those that anticipate and respond to the huge changes shaping people's lives across the world.

We are more certain than ever that it is the right time to focus our sustainability efforts on the innovative and eco-friendly products development.

Mission

Where to Play?

Our mission is to help our customers to produce high_quality and safe detergents that protect the health of consumers and environment through high quality, innovative and natural raw materials.

Core Value:

How do we do business?

Care: We act responsibly toward our customers, colleagues. Consumers, brand and our society and environment.

Simplicity: We strive for clarity and consistency, making decision quickly and pragmatically, and focusing on what is essential.

Trust: We say what we mean, keep our promises, and treat others with respect.

Transparency: We are transparent in ideas and thoughts and committed to bold objectives, take initiative, learn from mistakes, and see changes as an opportunity.

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Our Technology



Sulfated and sulfonated surfactants are industrially manufactured by reacting organic feedstocks (such as Linear Alkylbenzene, Primary Fatty Alcohols, and Ethoxylated Fatty Alcohols) with diluted gaseous SO_3.
The reaction is called sulfation or sulfonation, depending on the functional group formed in the product: sulfates **(R–O–SO_3H)** or sulfonates **(R–SO_3H)**. Sulfonation takes place in a **Multi-Tube Falling Film Reactor (MTFR) .**

The sulfonic acid obtained by processing alkylates (such as LABSA) can be marketed directly for industrial applications. Otherwise, the organic acids resulting from the sulfation reaction are immediately neutralized to form the corresponding organic salts such as **SLES, SLS** and **ALES**, due to minimizing by-product formation such as 1,4-dioxane.

To produce stable, homogeneous sulfated surfactants with the lowest levels of by-products and the desired specifications, **we use a Wiped Film Neutralizing System. This system allows the sulfation plant to operate at higher gas concentrations while maintaining dioxane levels below 50 ppm.**

The **Dioxane Removal System (DRS)** is a compact, low-energy neutralizing unit that removes volatile components from the SLES slurry under carefully controlled temperature and pressure conditions.
This ensures that the quality of heat-sensitive surfactants is preserved during the 1,4-dioxane removal process.



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The Flow of Value

Our Products

01 Page 07-08 — **Linear Alkylbenzene Sulfonic Acid** (LABSA 90% - LABSA 97%)

02 Page 09-10 — **Sodium Lauryl Ether Sulfate** (N30 – N70)

03 Page 13-14 — **Sodium Lauryl Sulfate** (SLS Liquid, Powder, Needle, Colored Needle)

04 Page 15-16 — **Amonium Lauryl Ether Sulfate** (ALES)

05 Page 17-18 — **Triethanolamine Lauryl Ether Sulfate** (TLES)

01 ASODEX 96 (LABSA)

Application

Linear Alkyl Benzene Sulfonic Acid (LABSA) is one of the most significant organosulfur compounds. It is extensively used as an anionic surfactant in detergent formulations and as an emulsifier in organic syntheses. Thanks to its excellent antimicrobial, wetting, cleaning and dispersing properties, it is employed in the agricultural sector for the production of pesticides and herbicides. In the textile industry, it functions as a wetting agent, emulsifier and dispersant in the manufacture of textile auxiliaries. Moreover, LABSA is a key component in the production of paper, leather and paint, and it plays a vital role as an emulsifier and foaming agent in the oil and gas industries.

Production

LABSA is produced through the reaction between Linear Alkylbenzene (LAB) and SO_3 gas. Gaseous SO_3 is produced by the reaction of molten sulfur in the presence of excess oxygen (forming SO_2 gas), followed by the reaction of the resulting SO_2 gas with oxygen in the presence of a catalyst.
In summary, the raw materials involved in the main process are LAB and Sulfur.

Available Packaging

Exir ASO Chemical Industries can supply ASODEX by 220 PE Drums, in Bulk and IBC Bulk packaging. It also depends on customer demands.

Analysis

Appearance	Amber viscous liquid
Active matter (%)	Min 96
Free oil (%)	Max 2
Mean molecular weight (g/mol)	322 ± 2
Sulfuric acid (%)	Max 1.8
Color (5% solution) (Klett)	Max 25
Acid Value (mg KOH/g sample)	178-190
Water (%)	Max 0.5
pH value (5% solution)	2-3

MSDS

- **Ingestion:** The product may cause severe irritation of the mouth and esophagus. If accidentally ingested, administer water. Do not induce vomiting. Call a physician.
- **Inhalation:** The product is not volatile. If under particular conditions, such as in the case of fire, fumes or aerosols are inhaled, remove the patient to a well-ventilated location and call a physician.
- **Eye Contact:** The product may cause burns. Flush eyes immediately with running water for a long time. Seek medical assistance immediately.
- **Skin Contact:** The product may cause burns. Remove the contaminated clothing and wash with plenty of water, and call a physician immediately.
- **Fire hazards:** LABSA may burn, but does not readily ignite. Poisonous gases such as sulfur oxides and hydrogen sulfide are produced in fire. If it is burned, use dry chemical, CO_2, or foam extinguishers.

02 Aponex 70 (SLES 70%)

Application

Sodium Lauryl Ether Sulfate (SLES) is an anionic surfactant that possesses good foaming and emulsifying properties. It demonstrates superior detergency, good tolerance to hard water, and mild irritation to sensitive skin and eyes, making it an ideal cleaning agent for household, personal care, and cosmetic products. Due to its good emulsifying characteristics, SLES is widely used as a wetting agent, as a surfactant for flooding in the oil industry, as an emulsifier in the dyeing and resin industries.

Analysis

Appearance	Clear/light yellow paste
Active Matter (%)	68-72
Mean molecular weight (g/mol)	384
1,4-Dioxane (ppm)	Max 35
Unsulfated Matter (%)	Max 2
Sodium Sulfate (%)	Max 1.5
Sodium Chloride (%)	Max 0.5
Fe (ppm)	Max 20
pH (10% solution)	7-9
Color (10% solution) (Hazen)	Max 20

Production

SLES is produced by the reaction between Ethoxylated Fatty Alcohol-2EO and SO_3 gas. Initially, molten sulfur is burned to form sulfur dioxide (SO_2), which is subsequently used for the production of SO_3/air.
To sum up, the raw materials used in the main process are Ethoxylated Fatty Alcohol-2EO, Sulfur, Caustic Soda and Soft Water.

Available Packaging

Exir ASO Chemical Industries offers SLES 70% in 60-110-150 L PE Drums, as well as IBC Bulk packaging, with options tailored to customer needs.

MSDS

- **Eye Contact:** May cause severe irritation to the eyes. Flush eyes immediately with running water for at least 15 minutes. Seek medical assistance immediately.
- **Skin Contact:** This product may cause severe irritation to the skin. In case of skin contact, remove contaminated clothing and wash with plenty of water.
- **Inhalation:** Do not breathe dust, fumes, gas, mist, vapors, or spray. Prolonged inhalation may be harmful.
- **Fire hazards:** Not flammable. Poisonous gases such as sulfur oxides and hydrogen sulfide are produced in fire. If it is burned, use dry chemical, CO_2, or foam extinguishers.
- **Maintenance situation:** Store in a cool and dry place away from heat and direct sunlight.

This is How we will grow our

BUSINESS

We strive to do better for our earth and our society. We want to act on the social and environmental issues facing the world, and we want to enhance people's lives with our innovative, sustainable and high-quality products.

Our Core Message:
We Act Responsibly

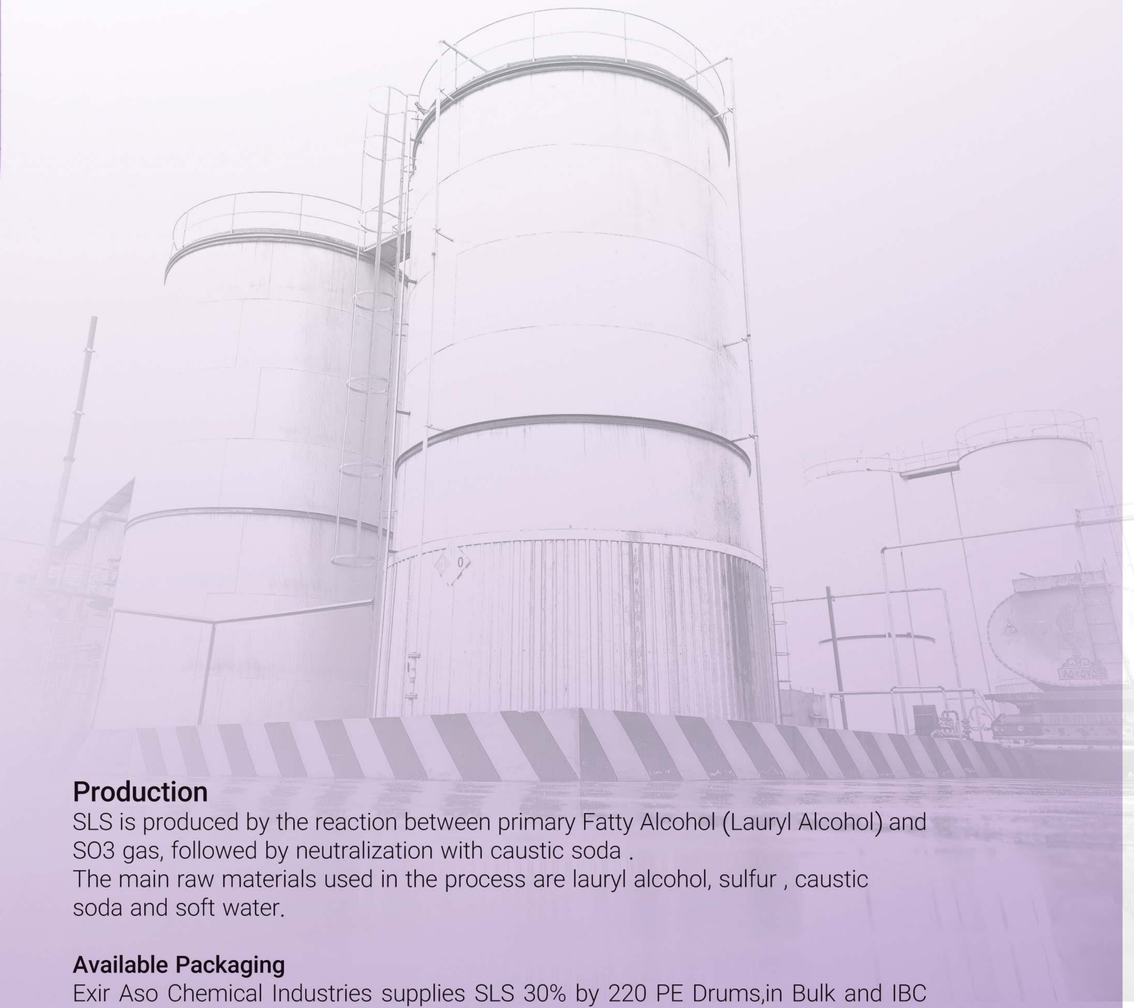
We act responsibly toward our customers, Colleagues, Consumers, Brand, Our Society and Environment.

03

SLS
(Sodium Lauryl Sulfate)

(SLS Liquid, Powder, Needle, Colored Needle)





Application
It is an anionic surfactant, similar to other surfactants such as SLES (Sodium Lauryl Ether Sulfate). It improves foam formation and increases cleaning power. Since this substance has antibacterial properties, it enhances the effect of preservatives and increases their effectiveness. It is used in the production of various types of soaps, toothpaste, surface cleaners, hair and body shampoos.

Analysis

SLS 30%

Appearance	colorless to pale yellow liquid
Active matter (%)	28-32
Mean Molecular mass (g/mol)	288.38
pH (10% w/w solution)	6-9
Unsulfated matter (%)	Max 1
Sodium Sulfate (%)	Max 1
Color (10% solution) (Hazen)	Max 20
Alkalinity (%)	Max 0.1

SLS 94%

Appearance	White granule, needle or powder
Active matter (%)	92-95
Mean Molecular weight (g/mol)	288.38
pH (10% w/w solution)	7.5-10.5
Unsulfated matter (%)	Max 1
Water Content (%)	Max 4
Sodium Sulfate (%)	Max 2

Production
SLS is produced by the reaction between primary Fatty Alcohol (Lauryl Alcohol) and SO3 gas, followed by neutralization with caustic soda .
The main raw materials used in the process are lauryl alcohol, sulfur , caustic soda and soft water.

Available Packaging
Exir Aso Chemical Industries supplies SLS 30% by 220 PE Drums,in Bulk and IBC Bulk packaging, and SLS 94% by 25 kg bag, depending on customer requirements.

MSDS
- **Inhalation:** Under particular conditions, do not inhale and move to a well-ventilated location.
- **Ingestion:** If the product is accidentally ingested, rinse mouth. Do not induce vomiting. Call a POISON CENTER/doctor if you feel unwell.
- **Ingestion:** If the product is accidentally ingested, rinse mouth. Do not induce vomiting. Call a POISON CENTER/doctor if you feel unwell.
- **Eye Contact:** Rinse cautiously with water for several minutes. Remove contact lenses if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/doctor.
- **Skin Contact:** If skin irritation occurs, get medical advice/attention. Flush skin with running water. Take off contaminated clothing and wash before reuse. Immediately call a POISON CENTER/doctor.
- **Fire Hazards:** It produces irritating or toxic gases in a fire. During ignition, extinguish with dry powder CO_2 and water vapor.
  Maintenance situation: Do not keep in a high-temperature place (50 °C) for more than one week.

04 ALES 70%
(Amonium Lauryl Ether Sulfate 70%)

Application
Amonium Lauryl Ether Sulfate is a mild anionic surfactant which is stable in hard water, and as a result, its detergency and foaming ability don't decrease. It has a large molecule; therefore, its penetration into the skin is lower than other surfactants, which makes it a suitable substance for baby products. It is widely used in liquid detergents, shampoos, gels, mild acidic shampoos, body shampoos, etc.

Production
ALES is the product of a reaction between ethoxylated fatty alcohol (ethoxylated with two moles of ethylene oxide) and SO_3 gas, followed by neutralization with ammonia.



Available Packaging
Exir ASO Chemical Industries can supply ALES in 60-110-150 PE Drums, and IBC Bulk packaging, depending on customer demands.

Analysis

Appearance	White or light-yellow paste
Active matter (%)	68-72
Mean Molecular weight (g/mol)	381
pH (5% w/w solution)	5-7
1,4-Dioxane (ppm)	Max 35
Unsulfated matter (%)	Max 3
Color (10% solution) (Hazen)	Max 20
Fe (ppm)	Max 20
Ammonium Sulfate (%)	Max 3
Free amine (%)	Max 0.05

MSDS
- **Ingestion:** If accidentally ingested, administer water. Do not induce vomiting.
- **Eye Contact:** Moderately irritating to the eyes. Flush eyes immediately with running water for 15 minutes. Seek medical assistance immediately.
- **Skin Contact:** Moderate skin irritation. In case of skin contact, remove contaminated clothing and wash with plenty of water.
- **Fire hazards:** During ignition, it produces irritating or toxic gases. Extinguish with dry chemical powder, foam, CO_2 and water vapor.
- **Maintenance situation:** Store in a dry and cool place, away from heat and sunlight.
- **Prohibitions:** Smoking is prohibited during transportation.

05

TLES 30

(Triethanolamine Lauryl Ether Sulfate 30%)

Application

Triethanolamine Lauryl Ether Sulfate is an anionic surfactant in the form of a transparent yellow liquid. Due to neutralization with triethanolamine, this compound has mild cleaning properties and is mainly used in the formulation of health care products, such as body and hair shampoos.



Analysis

Appearance	pale yellow viscous liquid
Active matter (%)	28-32
Mean molecular weight (g/mol)	514
pH (5% w/w solution)	6.5-7.5
1,4-Dioxane (ppm)	Max 35
Unsulfated matter (%)	Max 2.5
Color (10% solution) (Hazen)	Max 20
Fe (ppm)	Max 20
Triethanolamine Sulfate (%)	Max 1
Free amine (%)	Max 3

Production

TLES is the product of a reaction between ethoxylated fatty alcohol-2EO and SO_3 gas, followed by neutralization with triethanolamine.
Raw materials applied in the main process are ethoxylated fatty alcohol-2EO, sulfur and triethanolamine as the neutralizing agent and soft water.

Available Packaging

Exir ASO Chemical Industries can supply TLES 30% in 220 PE Drums, in Bulk and IBC bulk packaging. It also depends on customer demands.

MSDS

- **Ingestion:** Remove the person to fresh air and keep comfortable for breathing. If accidentally swallowed, administer water. Do not induce vomiting.
- **Ingestion:** If accidentally ingested the product, administer water. Do not induce vomiting.
- **Eye Contact:** Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/doctor.
- **Skin Contact:** May be harmful if absorbed through the skin. In case of skin contact, gently wash with plenty of soap and water.
- **Combustion:** During ignition, extinguish with dry powder CO2 and water vaper.

  Maintenance situation: Store in a dry and cool place away from heat and sunlight.
- **Fire Hazards:** It produces irritating or toxic gases in a fire. During ignition, extinguish with dry powder CO_2 and water vapor.
- **Maintenance situation:** Store in a dry and cool place, away from heat and sunlight.

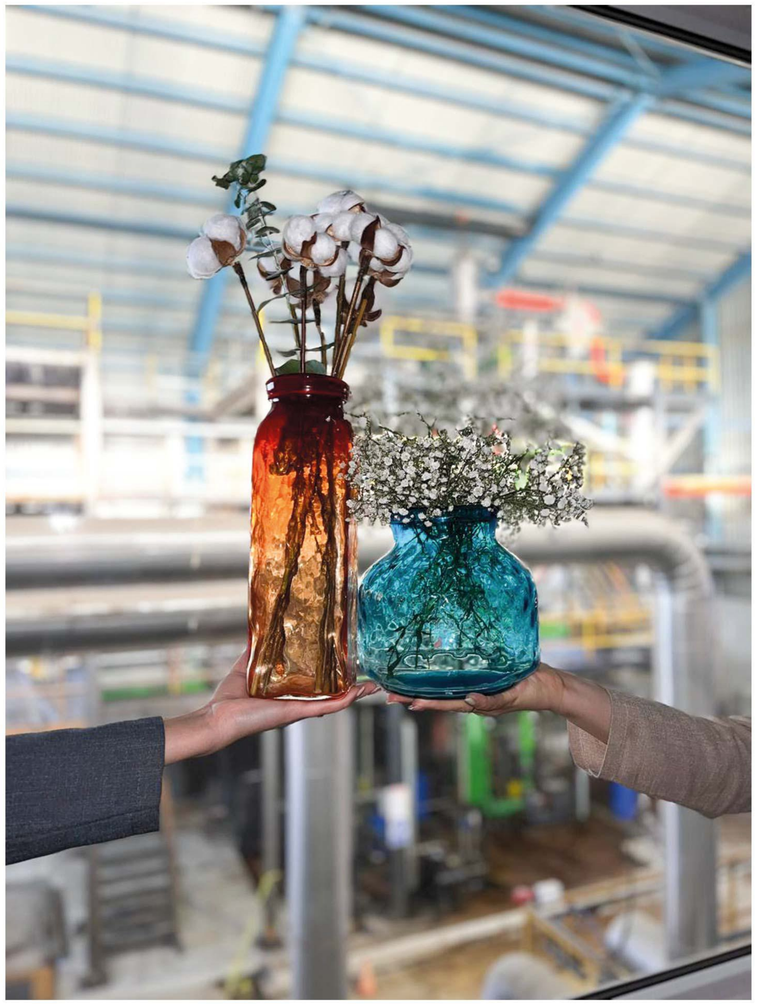


We're committed to contributing to the protection and regeneration of the environment, within and beyond our value chain.
The degradation of the earth's soil, increasing water scarcity and pollution of soil and air must be stopped and reversed for the health of the planet and society.

We are trying to make the world better

Protecting and regenerating nature takes a comprehensive approach. We're investing in infrastructure, technology and innovative products to prevent soil, air and water pollution due to detergent usage by people. We're also working to reduce emissions and preserve nature with developing of green area in our region.
Currently, we are working on programs such as creating green spaces inside and outside the factory complex, holding programs such as cleaning nature from waste, reducing the production of non-renewable waste and separating waste from the source and other programs to help the environment.
By working closely with others suppliers, industry partners, regulators and governments, we can deliver more impact. Alongside this, we're stepping up our advocacy for regulations that protect ecosystems and promote regenerative agriculture.

Quality Starts from your Decision



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CEO Message

Customers are the rulers of the market, and we are always available to the customers in the simplest and fastest ways to the satisfaction. In our organization, we try to have discipline with the least bureaucracy. Therefore, we adhere to the saying "Less is more." We expect everyone at ExirAso to be an ambassador for our high ethical standards. We want to create an environment where employees not only live our values in their own work-Care, Simplicity, Trust, and Transparency-but are vigilant in identifying potential concerns and confident about speaking up in such situations.
I am responsible to lead my colleagues to expand our business and leave a positive impact on our society during our working life.

Ribvar Nosrati
CEO